

Meta-Analysis Poolability Assessment

Health inequities in migrants with type 2 diabetes mellitus (T2DM)

Systematic review	Effect of healthcare inequities on complications in migrants with T2DM
Registration	PROSPERO CRD420261377625
Framework	Cochrane Handbook · PRISMA 2020 · SWiM (synthesis without meta-analysis)
Evidence base	13 studies · 118 extracted outcomes (Blocks D+E)
Scope	Poolability screening, effect-measure & unit harmonisation, pre-model flags

Executive summary

Row counts in the extraction table are misleading for synthesis. Three studies (Strooij = 36 rows, Consolazio = 10, Kristensen = 9) contribute most rows as *within-study subgroups* (ethnicity, region, sex, calendar year). When outcomes are counted by **distinct study**, only **8 of 23** clinical outcome groups reach the non-negotiable threshold of ≥3 studies — and several of those fail on clinical homogeneity, comparison direction, or confounding.

Two outcomes are defensibly meta-analysable: HbA1c glycaemic control (5 studies, **unanimous direction** — migrants consistently worse) and HbA1c monitoring / eye screening (process-of-care access). The hard clinical endpoints (composite CVD, all-cause mortality, stroke) are underpowered, confounded by the healthy-migrant effect, or compromised by within-migrant comparisons — these belong in **narrative / SWiM synthesis**, with the glycaemic-control pool as the quantitative centrepiece.

Cross-cutting exclusions applied to every pool

Study 7 (Okraïnec)	Compares immigrants <i>with vs. without language barrier</i> — a within-migrant contrast, not migrants vs. non-migrants. Fails Criterion 2 → excluded from all pools (eligible only for a separate language-barrier narrative strand).
Study 5 (Fiorini)	Undocumented migrants at an NGO clinic vs. Italian-born nationals; RR 0.16–0.32 for CVD (84–68% lower). Textbook healthy-migrant + undercount selection bias → flagged as an outlier; never pooled naively.
Unit conversions	None triggered. Every poolable outcome is a dimensionless ratio (OR/RR/HR) or proportion. Section 2 SI conversions apply only if continuous HbA1c% / glucose / lipid means are added later — absent from poolable groups here.

Poolability verdicts — all qualifying outcome groups

Outcome group	Distinct studies	Verdict
HbA1c glycaemic control (target / level)	5 (1, 6, 10, 12, 13)	CONDITIONAL — best candidate
HbA1c testing / monitoring	5 (1, 6, 10, 12, 13)	CONDITIONAL
Retinopathy / eye screening	4 (1, 6, 10, 11)	CONDITIONAL
CVD / composite CV events	5 → 3 after exclusions	CONDITIONAL (expect $I^2 > 75\%$)
All-cause mortality (T2DM) — POST-HOC	3 (3, 4, 8)	CONDITIONAL
Nephropathy / renal	3 (1, 5, 6)	NO — heterogeneous constructs
Retinopathy disease & treatment	3 (5, 6, 11)	NO — disease vs. procedures
Stroke	3 → 2 after S7 excluded	NO — < 3 studies
15 other groups	1–2	NO — < 3 studies → narrative

Detailed assessment — poolable / conditional groups

Each card follows the protocol output template (Section 5). Study IDs in parentheses; “use o##” refers to the extraction outcome_id to select per study.

HbA1c glycaemic control — % at / above target — strongest candidate		CONDITIONAL
Studies (n)	5 — S1, S6, S10, S12, S13	
Reason	Five studies, dichotomous-proportion family, unanimous direction (migrants worse glycaemic control). Measures convertible; P_0 available.	
Effect measure used	RR (primary)	
Conversions applied	OR→RR (Zhang & Yu) for S6 (OR 2.0) and S12 (OR 0.61–0.82). P_0 = comparator %-at-target is reported (~40–55%, not rare) → conversion required ; OR overestimates RR, especially S6 (OR > 2, $P_0 > 10\%$). S6 reports “HbA1c > 8%” = complement (poor control) → Section 4A inversion / declare valence.	
Direction verified	YES — all indicate worse control in migrants after valence harmonisation.	
Units verified	YES — dimensionless proportions (N/A).	
Primary model	Random-effects (DerSimonian–Laird), per protocol.	
Flags	Threshold heterogeneity ($\leq 7\%$ in S1/S12/S13 vs. $< 8\%$ in S10) → run as subgroup. Within-study multiplicity: S12 (4 ethnic), S13 (5 region), S10 (2 origin) must be combined to ONE estimate per study before pooling, or the reference group is double-counted and study weight inflated.	

HbA1c testing / monitoring (received guideline monitoring)		CONDITIONAL
Studies (n)	5 — S1, S6, S10, S12, S13	
Reason	≥3 studies, but the construct splits by monitoring frequency threshold.	
Effect measure used	RR	
Conversions applied	OR→RR for S12 (P_0 = comparator monitoring rate, high ~70–90% → conversion required). S6 “no HbA1c value reported” = complement → invert (Section 4A) to “monitored” before pooling.	
Direction verified	PARTIAL — most lower in migrants, but S12 Turkish/Moroccan OR > 1 (more monitoring) → genuine clinical heterogeneity by host country.	
Units verified	YES (N/A).	
Primary model	Random-effects (DerSimonian–Laird).	
Flags	Homogeneous poolable core = “≥1/yr monitoring” (S1, S10, S12 = 3 studies). ≥2/yr (S13) and ≥3/yr (S6) are different intensities → separate subgroups. Combine within-study subgroups first.	

Retinopathy / eye screening (received screening)		CONDITIONAL
Studies (n)	4 — S1, S6, S10, S11	
Reason	Four studies, consistent direction (all lower screening in migrants).	
Effect measure used	RR	
Conversions applied	OR→RR for S11 (Cox-derived OR; P_0 available).	
Direction verified	YES — S1 0.88, S6 0.86, S10 0.82/0.52, S11 0.76/0.77, all < 1.	
Units verified	YES (N/A).	
Primary model	Random-effects (DerSimonian–Laird).	
Flags	S1 (Adekunte) “Eye examination” 95% CI is corrupted in the source [0.887, 0.01196] — upper < lower, variance unusable → use point estimate only or drop S1 from inverse-variance weighting. Screening-interval heterogeneity (annual / 2-yr / 6-yr). Combine within-study subgroups (S10 ×2, S11 ×2 thresholds).	

Cardiovascular disease / composite CV events — high heterogeneity expected**CONDITIONAL**

Studies (n)	5 nominal → 3 usable (S2, S9, S10)
Reason	After excluding S7 (within-migrant comparison) and S5 (selection-bias outlier), 3 studies remain with divergent effects.
Effect measure used	Ratio measures (HR S2/S9 + RR S10) on the log scale
Conversions applied	None — HR/RR pooled together as ratio measures (Section 3C fallback); estimates are not all within 0.7–1.4, so justify cautiously.
Direction verified	NO — S2 1.38 (worse), S9 0.75–1.13 (mixed by sex), S10 1.22 (worse).
Units verified	YES (N/A).
Primary model	Random-effects (DerSimonian–Laird).
Flags	Use S2 o10 (all subjects) ONLY; drop sex/age subgroups o11–14. S9 reports sex-stratified only → combine to one estimate. Varying composite definitions and follow-up (3 yr vs. 6-yr cumulative). Likely $I^2 > 75\%$ → if so, do NOT interpret the pooled estimate; narrative synthesis per SWiM.

All-cause mortality (within T2DM) — POST-HOC (Amendment 1) — label post-hoc**CONDITIONAL**

Studies (n)	3 — S3, S4, S8
Reason	Three studies, all HR (compatible), all T2DM — but directions diverge sharply.
Effect measure used	HR
Conversions applied	None (all HR).
Direction verified	NO — S3 1.08 (NS), S4 1.42 (worse), S8 0.55–0.76 (better ; healthy-migrant / survivor effect, Sweden).
Units verified	YES (N/A).
Primary model	Random-effects (DerSimonian–Laird).
Flags	S3 — use o15 (first-generation Mexican-born) only; o16–19 are 2nd-generation US-born = not foreign-born migrants. S4 — use o23 (adjusted), not o22 (crude), per Priority Rule 1. S8 — combine 2 origin groups to one estimate. Mortality is confounding-sensitive (age structures differ sharply) + mixed adjustment sets → $I^2 > 75\%$ likely → narrative per SWiM. Must be labelled POST-HOC in the synthesis.

Not poolable — key reasons

Nephropathy / renal (3 studies)	Clinically different constructs: S1 normoalbuminuria (favourable result), S5 nephropathy diagnosis, S6 renal failure / ESRD. Mixed direction. Fails Criterion 3 → narrative.
Retinopathy disease & treatment (3 studies)	S5 / S6 = retinopathy disease & complications; S11 = surgical procedures (laser, vitrectomy). Disease prevalence ≠ procedure utilisation; three different measures (RR / OR / HR) → narrative.
Stroke	Drops to 2 studies once S7 is excluded; S9 / S10 stroke estimates are sparse and extreme (RR 0.14, CI to 3.17) → narrative.

15 single- / two-study groups

BP monitoring & target, LDL monitoring & target, lipid testing, kidney monitoring, foot exam, GP visits, hospital utilisation, glycosuria, diabetic foot, CHD, CV mortality, diabetes-specific mortality, acute complications: < 3 studies → narrative only (Rule 2). Most BP / LDL / foot / kidney rows are all Study 12 — high row counts, a single study.

Bottom line & recommended next steps

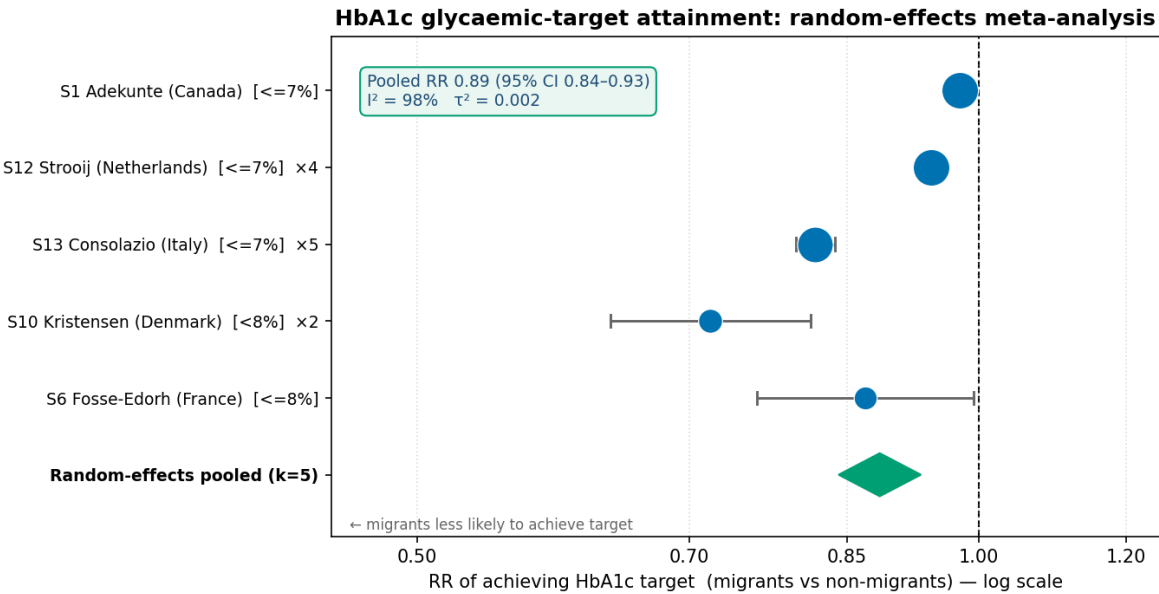
- **Quantitative core:** the HbA1c glycaemic-control pool (5 studies, unanimous direction) is the defensible headline meta-analysis; HbA1c monitoring and eye-screening pools support a process-of-care access narrative.
- **Narrative / SWiM:** composite CVD, all-cause mortality and stroke — underpowered, confounded by the healthy-migrant effect, or built on within-migrant comparisons.
- **Before modelling:** (1) apply OR→RR conversions (S6, S11, S12) using the reported comparator proportions as P_0 ; (2) collapse within-study subgroups to one estimate per study; (3) select adjusted over crude estimates (S4 o23, S3 o15); (4) compute I^2 and abandon pooled interpretation where $I^2 > 75\%$ (SWiM).
- **Declare every transformation** — no silent conversions (Priority Rule 3).

Quantitative synthesis — HbA1c glycaemic-target attainment

Worked random-effects meta-analysis of the headline pool. Outcome harmonised to the **relative risk (RR)** of a **migrant achieving the HbA1c target vs a non-migrant** (RR < 1 = migrants less likely to achieve control). Model: **DerSimonian–Laird random effects** on the log scale. All transformations are declared (Priority Rule 3).

Study (setting)	Thresh	Harmonised RR (95% CI)	Wt %	Harmonisation applied
S1 Adekunle (Canada)	<=7%	0.977 (0.970–0.984)	27.2	None — RR reported directly
S12 Strooij (Netherlands)	<=7%	0.943 (0.937–0.949)	27.2	OR→RR (Zhang–Yu, P ₀ = 0.83); 4 ethnic subgroups combined (fixed-effect)
S13 Consolazio (Italy)	<=7%	0.817 (0.798–0.837)	25.7	None — RR; 5 region subgroups combined (fixed-effect)
S10 Kristensen (Denmark)	<8%	0.718 (0.635–0.813)	10.5	None — RR; 2 origin subgroups combined (fixed-effect)
S6 Fosse-Edorh (France)	<=8%	0.870 (0.761–0.994)	9.5	OR(>8%) inverted → OR(≤8%) → RR (Zhang–Yu, P ₀ = 0.85)

Model	k	Pooled RR (95% CI)	I ²	τ ²
Primary — HbA1c ≤7% target	3	0.912 (0.863–0.963)	99.1%	0.0023
Sensitivity — all studies (mixed threshold)	5	0.885 (0.841–0.931)	98.3%	0.0025
Subgroup: ≤7% threshold	3	0.912 (0.863–0.963)	99.1%	0.0023
Subgroup: ~8% threshold	2	0.789 (0.654–0.951)	76.4%	0.0139



Interpretation & mandatory caveat (Priority Rule 5/7). Heterogeneity is extreme — I² = 98%, far above the 75% ceiling. Per SWiM, the pooled point estimate must **not** be reported as a single reliable effect. The robust, interpretable finding is the **unanimous direction**: in all five health systems migrants are less likely to achieve glycaemic control, with study-level RRs ranging ~0.72–0.98. The I² inflation is expected here — these are very large administrative cohorts (n up to millions), so within-study CIs are extremely tight and even modest, genuine between-setting differences in the disparity produce enormous Q. This is precise estimation of *different* effects, not noisy data. **Conclusion:** report as a consistent-direction disparity (narrative + this forest plot), not as a pooled RR headline.

Appendix — supporting figures

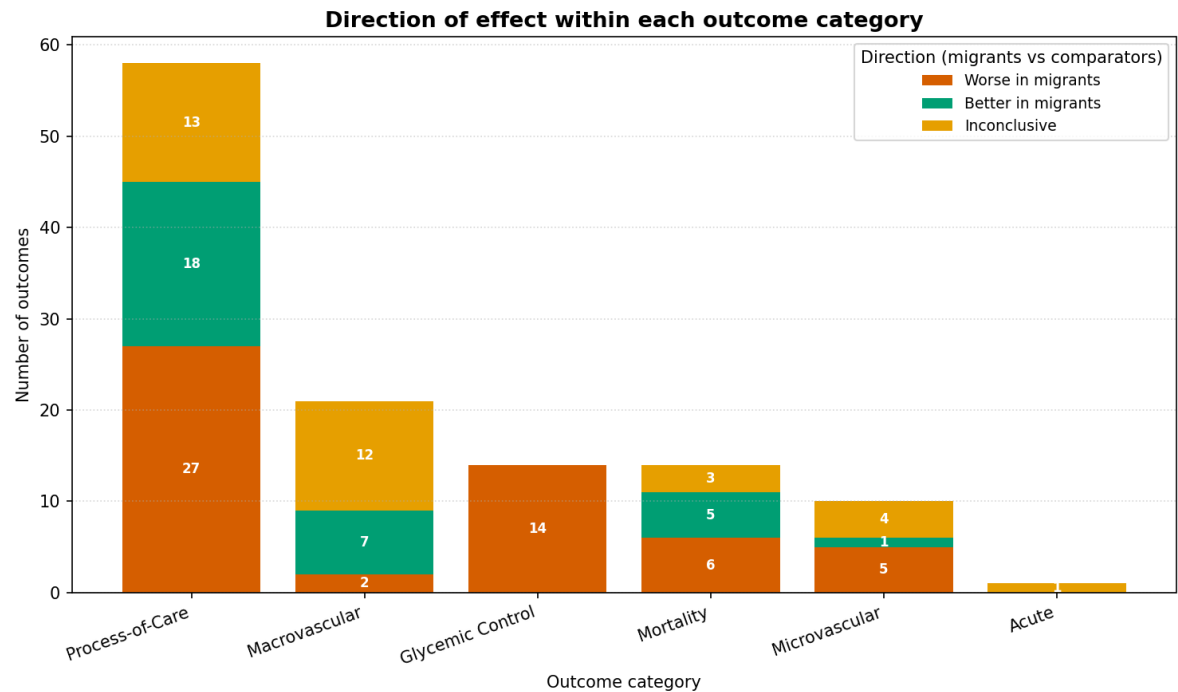


Figure 1. Direction of effect within each outcome category. The Glycemic Control bar is entirely “worse in migrants” (14/14) — the consistency underpinning the headline pool.

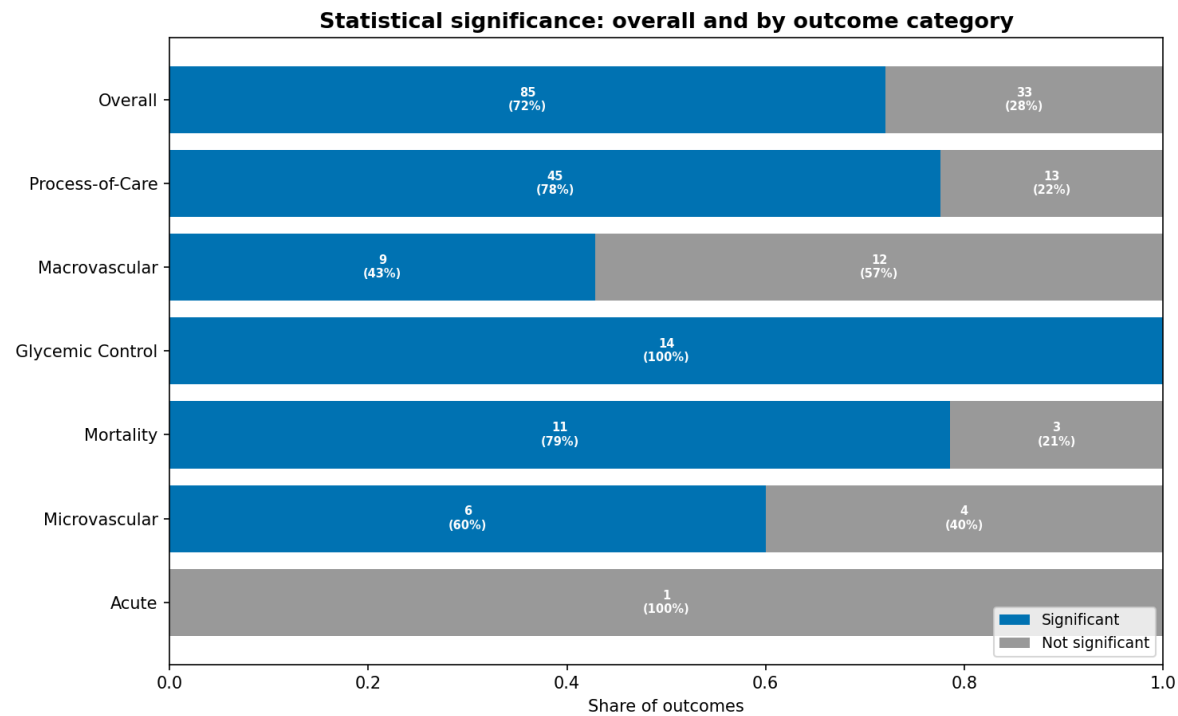


Figure 2. Statistical significance, overall and by category. Glycemic Control outcomes are 100% significant; macrovascular outcomes are only 43% significant (underpowered hard endpoints).